Note

The code for this chapter is in this book's GitHub repository in the chapter 8 folder: http://mng.bz/JzKp.

8.1. Tackling sparse rewards with predictive coding

In the world of neuroscience, and particularly computational neuroscience, there is a framework for understanding neural systems at a high level called the *predictive coding model*. In this model, the theory says that all neural systems from individual neurons up to large-scale neural networks are running an algorithm that attempts to predict inputs, and hence tries to minimize the *prediction error* between what it expects to experience and what it actually experiences. So at a high level, as you're going about your day, your brain is taking in a bunch of sensory information from the environment, and it's training to predict how the sensory information will evolve. It's trying to stay one step ahead of the actual raw data coming in.

If something surprising (i.e., unexpected) happens, your brain experiences a large prediction error and then presumably does some parameter updating to prevent that from happening again. For example, you might be talking to someone you just met, and your brain is constantly trying to predict the next word that person will say before they say it. Since this is someone you don't know, your brain will probably have a relatively high average prediction error, but if you become best friends, you'll probably be quite good at finishing their sentences. This is not something you try to do; whether you want to or not, your brain is trying to reduce its prediction error.

Curiosity can be thought of as a kind of desire to reduce the uncertainty in your environment (and hence reduce prediction errors). If you were a software engineer and you saw some online posts about this interesting field called machine learning, your curiosity to read a book like this would be based on the goal of reducing your uncertainty about machine learning.